

Bookwork

2.1

1. The keyword to start a data structure is "struct".
2. The final punctuation to finish a data structure definition is a semicolon.
3. The symbol for a pointer to a data structure to access a member field is $\rightarrow$.
4. The default starting position for an enumerated type is 0.
5. an enumerated data type can be used for the index of an array.

2.2

①

```
struct Cave
{
    int stalactites;
    int stalagmites;
    int pits;
    String name;
    double lon;
    double lat;
};
```

②

```
enum colors {Red, Yellow, Green, Cyan, Blue, Magenta};
struct ColorNode
{
    int redbird;
    int bluebird;
    int greenbird;
};

int main() {
    ColorNode array[6];
    For (int i = 0; i < 6; i++)
    {
        array[i].redbird = 0;
        array[i].bluebird = 0;
        array[i].greenbird = 0;
    }

    array[Red].redbird = 1;
    array[Yellow].redbird = 1;
    array[Magenta].redbird = 1;
    array[Cyan].bluebird = 1;
    array[Blue].bluebird = 1;
    array[Magenta].bluebird = 1;
    array[Yellow].greenbird = 1;
    array[Green].greenbird = 1;
    array[Cyan].greenbird = 1;
}
```

Continued

③

```
int main ( ) {  
    enum color { Red, Yellow, Green, Cyan, Blue, Magenta };  
    int redfish[6] = { 0, 0, 0, 0, 0, 0 };  
    int bluefish[6] = { 0, 0, 0, 0, 0, 0 };  
    int greenfish[6] = { 0, 0, 0, 0, 0, 0 };  
    redfish[Red] = 1;  
    redfish[Yellow] = 1;  
    redfish[Magenta] = 1;  
    bluefish[Cyan] = 1;  
    bluefish[Blue] = 1;  
    bluefish[Magenta] = 1;  
    greenfish[Yellow] = 1;  
    greenfish[Green] = 1;  
    greenfish[Cyan] = 1;  
}
```

3

④

```
struct Node  
{  
    int Key;  
    string Payload;  
    struct Node *Next;  
};
```

[illegible]